9. Convert lowercase characters to uppercase

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]>='a'&&s[i]<='z'){
         s[i]=s[i]-32;}}
   printf("Result = %s",s);}
```

10. Convert uppercase characters to lowercase

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]>='A'&&s[i]<='Z'){
         s[i]=s[i]+32;}}
   printf("Result = %s",s);}
```

11. Toggle case of every character

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]>='a'&&s[i]<='z'){
         s[i]=s[i]-32;}
      else if(s[i]>='A'&&s[i]<='Z'){
         s[i]=s[i]+32;}}
   printf("Result = %s",s);}
```

12. Print each character on a separate line

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]!='\n'){
         printf("%c\n",s[i]);}}}
```

13. Print ASCII value of every character

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      printf("%c = %d\n",s[i],s[i]);}}
```

14. Count number of alphabets

```
#include <stdio.h>
int main(){
   char s[100];
   int i,count=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
if((s[i]>='a'&&s[i]<='z')||(s[i]>='A'&&s[i]<='Z'))
{
         count++;}}
   printf("Alphabets = %d",count);}
```

15. Count special characters

```
#include <stdio.h>
int main(){
   char s[100];
   int i,count=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(!((s[i]>='a'&&s[i]<='z')||
         (s[i]>='A'&&s[i]<='Z')||
         (s[i]>='0'&&s[i]<='9')||
         s[i]==' '||s[i]=='\n')){
         count++;}}
   printf("Special characters = %d",count);}
```

16. Find first character

```
#include <stdio.h>
int main(){
   char s[100];
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("First character = %c",s[0]);}
```

17. Find last character

```
#include <stdio.h>
int main(){
   char s[100];
   int i=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   while(s[i]!='\0'&&s[i]!='\n'){
```

```
    i++;}
  printf("Last character = %c",s[i-1]);}
```

18. Copy string without strcpy()

```
#include <stdio.h>
int main(){
  char s[100],copy[100];
  int i;
  printf("Enter string: ");
  fgets(s,100,stdin);
  for(i=0;s[i]!='\0';i++){
    copy[i]=s[i];}
  copy[i]='\0';
  printf("Copied string = %s",copy);}
```

19. Compare two strings without strcmp()

```
#include <stdio.h>
int main(){
  char a[100],b[100];
  int i=0,same=1;
  printf("Enter first string: ");
  fgets(a,100,stdin);
  printf("Enter second string: ");
  fgets(b,100,stdin);
  while(a[i]!='\0'&&b[i]!='\0'){
    if(a[i]=='\n'||b[i]=='\n'){
      break;}
    if(a[i]!=b[i]){
      same=0;
      break;}
    i++;}
if(same&&a[i]!=b[i]&&a[i]!='\n'&&b[i]!='\n'){
    same=0;}
  if(same){
    printf("Strings are equal");}
  else{
    printf("Strings are different");}}
```

20. Check whether two strings are exactly equal

```
#include <stdio.h>
int main(){
  char a[100],b[100];
  int i=0;
  printf("Enter first string: ");
  fgets(a,100,stdin);
  printf("Enter second string: ");
  fgets(b,100,stdin);
  while(a[i]==b[i]&&a[i]!='\0'){
    i++;}
  if(a[i]==b[i]){
    printf("Exactly equal");}
  else{
    printf("Not equal");}}
```

LEVEL 2: STRING MANIPULATION

21. Reverse a string without strrev()

```
#include <stdio.h>
int main(){
  char s[100];
  int i=0,j;
  char temp;
  printf("Enter string: ");
  fgets(s,100,stdin);
  while(s[i]!='\0'&&s[i]!='\n'){
    i++;}
  j=i-1;
  for(i=0;i<j;i++,j--){
    temp=s[i];
    s[i]=s[j];
    s[j]=temp;}
  printf("Reverse = %s",s);}
```

22. Reverse a string using a loop

```
#include <stdio.h>
int main(){
  char s[100];
  int i=0;
  printf("Enter string: ");
  fgets(s,100,stdin);
  while(s[i]!='\0'&&s[i]!='\n'){
    i++;}
  printf("Reverse = ");
  for(i=i-1;i>=0;i--){
    printf("%c",s[i]);}}
```

23. Reverse a string using pointers

```
#include <stdio.h>
int main(){
  char s[100];
  char *p;
  int i=0;
  printf("Enter string: ");
  fgets(s,100,stdin);
  while(s[i]!='\0'&&s[i]!='\n'){
    i++;}
  p=s+i-1;
  printf("Reverse = ");
  while(p>=s){
    printf("%c",*p);
    p--;}}
```

24. Reverse each word in a sentence

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,end;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]==' '||s[i]=='\n'){
         end=i-1;
         while(start<end){
            char temp=s[start];
            s[start]=s[end];
            s[end]=temp;
            start++;
            end--;}
         start=i+1;}
      i++;}
   printf("Result = %s",s);}
```

25. Reverse order of words in a sentence

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200];
   char words[20][30];
   int i=0,j=0,k=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'&&s[i]!='\n'){
      if(s[i]!=' '){
         words[j][k]=s[i];
         k++;}
      else{
         words[j][k]='\0';
         j++;
         k=0;}
      i++;}
   words[j][k]='\0';
   printf("Reverse words = ");
   for(i=j;i>=0;i--){
      printf("%s ",words[i]);}}
```

26. Concatenate two strings without strcat()

```
#include <stdio.h>
int main(){
   char a[200],b[100];
   int i=0,j=0;
   printf("Enter first string: ");
   fgets(a,100,stdin);
   printf("Enter second string: ");
   fgets(b,100,stdin);
   while(a[i]!='\0'&&a[i]!='\n'){
      i++;}
   while(b[j]!='\0'&&b[j]!='\n'){
      a[i]=b[j];
      i++;
      j++;}
   a[i]='\0';
   printf("Result = %s",a);}
```

27. Insert one string into another at a given position

```
#include <stdio.h>
int main(){
   char a[200],b[100];
   int pos,i,j=0,len=0;
   printf("Enter first string: ");
   fgets(a,100,stdin);
   printf("Enter string to insert: ");
   fgets(b,100,stdin);
   printf("Enter position: ");
   scanf("%d",&pos);
   while(a[len]!='\0'&&a[len]!='\n'){
      len++;}
   if(pos<0||pos>len){
      printf("Invalid position");
      return 0;}
   for(i=len;i>=pos;i--){
      a[i+(len+1)]=a[i];}
   while(b[j]!='\0'&&b[j]!='\n'){
      a[pos]=b[j];
      pos++;
      j++;}
   printf("Result = %s",a);}
```

28. Delete a character from a given position

```
#include <stdio.h>
int main(){
   char s[100];
   int pos,i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter position: ");
   scanf("%d",&pos);
   for(i=pos;s[i]!='\0';i++){
      s[i]=s[i+1];}
   printf("Result = %s",s);}
```

29. Delete all occurrences of a particular character

```
#include <stdio.h>
```

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hiiiiiiii

```
int main(){
   char s[100],ch;
   int i,j=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter character: ");
   scanf(" %c",&ch);
   for(i=0;s[i]!='\0';i++){
      if(s[i]!=ch){
         s[j]=s[i];
         j++;}}
   s[j]='\0';
   printf("Result = %s",s);}
```

30. Replace a particular character with another

```
#include <stdio.h>
int main(){
   char s[100],a,b;
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter character to replace: ");
   scanf(" %c",&a);
   printf("Enter new character: ");
   scanf(" %c",&b);
   for(i=0;s[i]!='\0';i++){
      if(s[i]==a){
         s[i]=b;}}
   printf("Result = %s",s);}
```

31. Replace all vowels with *

```
#include <stdio.h>
int main(){
   char s[100];
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
if(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]=
='u'||
s[i]=='A'||s[i]=='E'||s[i]=='I'||s[i]=='O'||s[i]=='
U'){
         s[i]='*';}}
   printf("Result = %s",s);}
```

32. Remove all spaces

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]!=' '){
         s[j]=s[i];
         j++;}}
   s[j]='\0';
   printf("Result = %s",s);}
```

33. Remove all special characters

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if((s[i]>='a'&&s[i]<='z')||
        (s[i]>='A'&&s[i]<='Z')||
        (s[i]>='0'&&s[i]<='9')||
        s[i]==' '){
         s[j]=s[i];
         j++;}}
   s[j]='\0';
   printf("Result = %s",s);}
```

34. Remove duplicate characters

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j,k,found;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]=='\n'){
         s[i]='\0';
         break;}
      found=0;
      for(j=0;j<i;j++){
         if(s[i]==s[j]){
            found=1;
            break;}}
      if(found){
         for(k=i;s[k]!='\0';k++){
            s[k]=s[k+1];}
         i--;}}
   printf("Result = %s",s);}
```

35. Remove duplicate words from a sentence

```
#include <stdio.h>
#include <string.h>
```

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hiiiiiiii
hiiiiiiii
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```
int main(){
   char s[200];
   char words[30][30];
   int i=0,j=0,k=0,n=0,found;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      found=0;
      for(i=0;i<n;i++){
         if(strcmp(words[i],p)==0){
            found=1;
            break;}}
      if(!found){
         strcpy(words[n],p);
         n++;}
      p=strtok(NULL," \n");}
   printf("Result = ");
   for(i=0;i<n;i++){
      printf("%s ",words[i]);}}
```

36. Find first occurrence of a character

```
#include <stdio.h>
int main(){
   char s[100],ch;
   int i,pos=-1;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter character: ");
   scanf(" %c",&ch);
   for(i=0;s[i]!='\0';i++){
      if(s[i]==ch){
         pos=i;
         break;}}
   if(pos==-1){
      printf("Character not found");}
   else{
      printf("First occurrence = %d",pos);}}
```

37. Find last occurrence of a character

```
#include <stdio.h>
int main(){
   char s[100],ch;
   int i,pos=-1;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter character: ");
   scanf(" %c",&ch);
   for(i=0;s[i]!='\0';i++){
      if(s[i]==ch){
         pos=i;}}
   if(pos==-1){
      printf("Character not found");}
   else{
      printf("Last occurrence = %d",pos);}}
```

38. Find number of occurrences of a character

```
#include <stdio.h>
int main(){
   char s[100],ch;
   int i,count=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   printf("Enter character: ");
   scanf(" %c",&ch);
   for(i=0;s[i]!='\0';i++){
      if(s[i]==ch){
         count++;}}
   printf("Occurrences = %d",count);}
```

39. Find number of occurrences of a word

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],word[30];
   int count=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   printf("Enter word: ");
   scanf("%s",word);
   char *p=strtok(s," \n");
   while(p!=NULL){
      if(strcmp(p,word)==0){
         count++;}
      p=strtok(NULL," \n");}
   printf("Occurrences = %d",count);}
```

40. Check whether a string contains only alphabets

```
#include <stdio.h>
int main(){
   char s[100];
   int i,valid=1;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(!((s[i]>='a'&&s[i]<='z')||
          (s[i]>='A'&&s[i]<='Z'))){
         valid=0;
         break;}}
   if(valid){
```

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```
    printf("Only alphabets");}
  else{
    printf("Contains other characters");}}
```

LEVEL 3: STRING LOGIC & PATTERNS

41. Check whether a string is palindrome

```
#include <stdio.h>
int main(){
  char s[100];
  int i=0,j,flag=1;
  printf("Enter string: ");
  fgets(s,100,stdin);
  while(s[i]!='\0'&&s[i]!='\n'){
    i++;}
  j=i-1;
  for(i=0;i<j;i++,j--){
    if(s[i]!=s[j]){
      flag=0;
      break;}}
  if(flag){
    printf("Palindrome");}
  else{
    printf("Not palindrome");}}
```

42. Palindrome ignoring spaces

```
#include <stdio.h>
int main(){
  char s[100],a[100];
  int i,j=0,k,flag=1;
  printf("Enter string: ");
  fgets(s,100,stdin);
  for(i=0;s[i]!='\0';i++){
    if(s[i]!=' '&&s[i]!='\n'){
      a[j]=s[i];
      j++;}}
  a[j]='\0';
  k=j-1;
  for(i=0;i<k;i++,k--){
    if(a[i]!=a[k]){
      flag=0;
      break;}}
  if(flag){
    printf("Palindrome");}
  else{
    printf("Not palindrome");}}
```

43. Palindrome ignoring case

```
#include <stdio.h>
int main(){
  char s[100];
  int i=0,j,flag=1;
  printf("Enter string: ");
  fgets(s,100,stdin);
  while(s[i]!='\0'&&s[i]!='\n'){
    i++;}
  j=i-1;
  for(i=0;i<j;i++,j--){
    char a=s[i],b=s[j];
    if(a>='A'&&a<='Z'){
      a=a+32;}
    if(b>='A'&&b<='Z'){
      b=b+32;}
    if(a!=b){
      flag=0;
      break;}}
  if(flag){
    printf("Palindrome");}
  else{
    printf("Not palindrome");}}
```

44. Check whether two strings are anagrams

```
#include <stdio.h>
int main(){
  char a[100],b[100];
  int count[256]={0};
  int i,flag=1;
  printf("Enter first string: ");
  fgets(a,100,stdin);
  printf("Enter second string: ");
  fgets(b,100,stdin);
  for(i=0;a[i]!='\0';i++){
    if(a[i]!=' '&&a[i]!='\n'){
      count[(unsigned char)a[i]]++;}}
  for(i=0;b[i]!='\0';i++){
    if(b[i]!=' '&&b[i]!='\n'){
      count[(unsigned char)b[i]]--;}}
  for(i=0;i<256;i++){
    if(count[i]!=0){
      flag=0;
      break;}}
  if(flag){
    printf("Anagrams");}
  else{
    printf("Not anagrams");}}
```

45. Anagrams without sorting

```
#include <stdio.h>
int main(){
  char a[100],b[100];
  int count[26]={0};
  int i,flag=1;
```

```
   printf("Enter first string: ");
   fgets(a,100,stdin);
   printf("Enter second string: ");
   fgets(b,100,stdin);
   for(i=0;a[i]!='\0';i++){
      if(a[i]>='A'&&a[i]<='Z'){
         a[i]=a[i]+32;}
      if(a[i]>='a'&&a[i]<='z'){
         count[a[i]-'a']++;}}
   for(i=0;b[i]!='\0';i++){
      if(b[i]>='A'&&b[i]<='Z'){
         b[i]=b[i]+32;}
      if(b[i]>='a'&&b[i]<='z'){
         count[b[i]-'a']--;}}
   for(i=0;i<26;i++){
      if(count[i]!=0){
         flag=0;
         break;}}
   if(flag){
      printf("Anagrams");}
   else{
      printf("Not anagrams");}}
```

46. Find first non-repeated character

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i,found=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]==1){
         printf("First non-repeated = %c",s[i]);
         found=1;
         break;}}
   if(!found){
      printf("No non-repeated character");}}
```

47. Find first repeated character

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]>1){
         printf("First repeated = %c",s[i]);
         break;}}
```

48. Find all repeated characters

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   printf("Repeated characters = ");
   for(i=0;i<256;i++){
      if(count[i]>1){
         printf("%c ",i);}}
}
```

49. Find all non-repeated characters

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   printf("Non-repeated characters = ");
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]==1){
         printf("%c ",s[i]);}}
}
```

50. Find most frequently occurring character

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i,max=0;
   char ch;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]>max){
```

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```
        max=count[(unsigned char)s[i]];
        ch=s[i];}}
   printf("Most frequent = %c",ch);}
```

51. Find least frequently occurring character

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i,min=100000;
   char ch;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]<min){
         min=count[(unsigned char)s[i]];
         ch=s[i];}}
   printf("Least frequent = %c",ch);}
```

52. Find frequency of every character

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;i<256;i++){
      if(count[i]>0){
         printf("%c = %d\n",i,count[i]);}}
}
```

53. Find longest word in a sentence

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,len=0,max=0,maxstart=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>max){
            max=len;
            maxstart=start;}
         len=0;
         start=i+1;}
      i++;}
   printf("Longest word = ");
   for(i=maxstart;i<maxstart+max;i++){
      printf("%c",s[i]);}}
```

54. Find shortest word

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,len=0,min=1000,minstart=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>0&&len<min){
            min=len;
            minstart=start;}
         len=0;
         start=i+1;}
      i++;}
   printf("Shortest word = ");
   for(i=minstart;i<minstart+min;i++){
      printf("%c",s[i]);}}
```

55. Find number of words

```
#include <stdio.h>
int main(){
   char s[200];
   int i,count=0,inword=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   for(i=0;s[i]!='\0';i++){
      if(s[i]!=' '&&s[i]!='\n'){
         if(inword==0){
            count++;
            inword=1;}}
      else{
         inword=0;}}
   printf("Words = %d",count);}
```

56. Find average length of words

```
#include <stdio.h>
int main(){
   char s[200];
   int i,len=0,words=0,total=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   for(i=0;s[i]!='\0';i++){
```

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```
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>0){
            total=total+len;
            words++;
            len=0;}}}
   if(len>0){
      total=total+len;
      words++;}
   printf("Average length =
%.2f",(float)total/words);}
```

57. Find word with maximum number of characters

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,len=0,max=0,maxstart=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>max){
            max=len;
            maxstart=start;}
         len=0;
         start=i+1;}
      i++;}
   printf("Maximum word = ");
   for(i=maxstart;i<maxstart+max;i++){
      printf("%c",s[i]);}}
```

58. Find word with minimum number of characters

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,len=0,min=1000,minstart=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>0&&len<min){
            min=len;
            minstart=start;}
         len=0;
         start=i+1;}
      i++;}
   printf("Minimum word = ");
   for(i=minstart;i<minstart+min;i++){
      printf("%c",s[i]);}}
```

59. Find second longest word

```
#include <stdio.h>
int main(){
   char s[200];
   int i=0,start=0,len=0,max=0,second=0;
   int maxstart=0,secondstart=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   while(s[i]!='\0'){
      if(s[i]!=' '&&s[i]!='\n'){
         len++;}
      else{
         if(len>max){
            second=max;
            secondstart=maxstart;
            max=len;
            maxstart=start;}
         else if(len>second&&len<max){
            second=len;
            secondstart=start;}
         len=0;
         start=i+1;}
      i++;}
   if(second>0){
      printf("Second longest word = ");
for(i=secondstart;i<secondstart+second;i++){
         printf("%c",s[i]);}}
   else{
      printf("Second longest word not
found");}}
```

60. Sort characters ascending

```
#include <stdio.h>
int main(){
   char s[100],temp;
   int i,j;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      for(j=i+1;s[j]!='\0'&&s[j]!='\n';j++){
         if(s[i]>s[j]){
            temp=s[i];
            s[i]=s[j];
```

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```
            s[j]=temp;}}}
   printf("Sorted = %s",s);}
```

LEVEL 4: INTERVIEW-LEVEL STRINGS

61. Sort characters descending

```
#include <stdio.h>
int main(){
   char s[100],temp;
   int i,j;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      for(j=i+1;s[j]!='\0'&&s[j]!='\n';j++){
         if(s[i]<s[j]){
            temp=s[i];
            s[i]=s[j];
            s[j]=temp;}}}
   printf("Sorted = %s",s);}
```

62. Find character with second-highest frequency

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i,first=0,second=0;
   char ch=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;i<256;i++){
      if(count[i]>first){
         second=first;
         first=count[i];}
      else if(count[i]>second&&count[i]<first){
         second=count[i];}}
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      if(count[(unsigned char)s[i]]==second){
         ch=s[i];
         break;}}
   if(ch!=0){
      printf("Second highest frequency =
%c",ch);}
   else{
      printf("Not found");}}
```

63. Find character with highest frequency without array

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j,count,max=0;
   char ch=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count=0;
      for(j=0;s[j]!='\0'&&s[j]!='\n';j++){
         if(s[i]==s[j]){
            count++;}}
      if(count>max){
         max=count;
         ch=s[i];}}
   printf("Highest frequency = %c",ch);}
```

64. Find duplicate words

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200];
   char words[30][30];
   int n=0,i,j,found;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      strcpy(words[n],p);
      n++;
      p=strtok(NULL," \n");}
   printf("Duplicate words = ");
   for(i=0;i<n;i++){
      found=0;
      for(j=0;j<i;j++){
         if(strcmp(words[i],words[j])==0){
            found=1;
            break;}}
      if(found){
         printf("%s ",words[i]);}}
}
```

65. Count frequency of every word

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200];
   char words[30][30];
   int count[30]={0};
   int n=0,i,j,found;
   printf("Enter sentence: ");
```

```
    fgets(s,200,stdin);
    char *p=strtok(s," \n");
    while(p!=NULL){
       found=-1;
       for(i=0;i<n;i++){
          if(strcmp(words[i],p)==0){
             found=i;
             break;}}
       if(found==-1){
          strcpy(words[n],p);
          count[n]=1;
          n++;}
       else{
          count[found]++;}
       p=strtok(NULL," \n");}
    for(i=0;i<n;i++){
       printf("%s = %d\n",words[i],count[i]);}}
```

66. Remove duplicate words preserving original order

```
#include <stdio.h>
#include <string.h>
int main(){
    char s[200];
    char words[30][30];
    int n=0,i,found;
    printf("Enter sentence: ");
    fgets(s,200,stdin);
    char *p=strtok(s," \n");
    while(p!=NULL){
       found=0;
       for(i=0;i<n;i++){
          if(strcmp(words[i],p)==0){
             found=1;
             break;}}
       if(!found){
          strcpy(words[n],p);
          n++;}
       p=strtok(NULL," \n");}
    printf("Result = ");
    for(i=0;i<n;i++){
       printf("%s ",words[i]);}}
```

67. Find longest palindrome substring

```
#include <stdio.h>
int main(){
    char s[100];
    int i,j,k,flag,max=0,start=0;
    printf("Enter string: ");
    fgets(s,100,stdin);
    for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
       for(j=i;s[j]!='\0'&&s[j]!='\n';j++){
          flag=1;
          for(k=i;k<=(i+j)/2;k++){
             if(s[k]!=s[j-k+i]){
                flag=0;
                break;}}
          if(flag&&j-i+1>max){
             max=j-i+1;
             start=i;}}}
    printf("Longest palindrome = ");
    for(i=start;i<start+max;i++){
       printf("%c",s[i]);}}
```

68. Find longest word having all unique characters

```
#include <stdio.h>
int main(){
    char s[200];
    int i=0,start=0,len=0,max=0,maxstart=0;
    printf("Enter sentence: ");
    fgets(s,200,stdin);
    while(s[i]!='\0'){
       if(s[i]!=' '&&s[i]!='\n'){
          len++;}
       else{
          int j,k,unique=1;
          for(j=start;j<start+len;j++){
             for(k=j+1;k<start+len;k++){
                if(s[j]==s[k]){
                   unique=0;}}}
          if(unique&&len>max){
             max=len;
             maxstart=start;}
          len=0;
          start=i+1;}
       i++;}
    printf("Longest unique word = ");
    for(i=maxstart;i<maxstart+max;i++){
       printf("%c",s[i]);}}
```

69. Check whether string contains all unique characters

```
#include <stdio.h>
int main(){
    char s[100];
    int i,j,unique=1;
    printf("Enter string: ");
    fgets(s,100,stdin);
    for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
```

```
    for(j=i+1;s[j]!='\0'&&s[j]!='\n';j++){
      if(s[i]==s[j]){
        unique=0;
        break;}}}
  if(unique){
    printf("All characters are unique");}
  else{
    printf("Duplicate characters found");}}
```

70. Check whether one string is rotation of another

```
#include <stdio.h>
#include <string.h>
int main(){
  char a[100],b[100],c[200];
  printf("Enter first string: ");
  fgets(a,100,stdin);
  printf("Enter second string: ");
  fgets(b,100,stdin);
  a[strcspn(a,"\n")]='\0';
  b[strcspn(b,"\n")]='\0';
  if(strlen(a)!=strlen(b)){
    printf("Not rotation");
    return 0;}
  strcpy(c,a);
  strcat(c,a);
  if(strstr(c,b)!=NULL){
    printf("Rotation");}
  else{
    printf("Not rotation");}}
```

71. Check whether one string is substring of another without strstr()

```
#include <stdio.h>
int main(){
  char a[100],b[50];
  int i,j,found=0;
  printf("Enter main string: ");
  fgets(a,100,stdin);
  printf("Enter substring: ");
  fgets(b,50,stdin);
  for(i=0;a[i]!='\0'&&a[i]!='\n';i++){
    j=0;
while(b[j]!='\0'&&b[j]!='\n'&&a[i+j]==b[j]){
      j++;}
    if(b[j]=='\0'||b[j]=='\n'){
      found=1;
      break;}}
  if(found){
    printf("Substring found");}
  else{
    printf("Substring not found");}}
```

72. Find all occurrences of substring

```
#include <stdio.h>
int main(){
  char a[200],b[50];
  int i,j,found;
  printf("Enter main string: ");
  fgets(a,200,stdin);
  printf("Enter substring: ");
  fgets(b,50,stdin);
  printf("Positions = ");
  for(i=0;a[i]!='\0'&&a[i]!='\n';i++){
    found=1;
    for(j=0;b[j]!='\0'&&b[j]!='\n';j++){
      if(a[i+j]!=b[j]){
        found=0;
        break;}}
    if(found){
      printf("%d ",i);}}}
```

73. Replace a word in a sentence with another word

```
#include <stdio.h>
#include <string.h>
int main(){
  char s[200],old[30],newword[30];
  char result[300]="";
  printf("Enter sentence: ");
  fgets(s,200,stdin);
  printf("Enter old word: ");
  scanf("%s",old);
  printf("Enter new word: ");
  scanf("%s",newword);
  char *p=strtok(s," \n");
  while(p!=NULL){
    if(strcmp(p,old)==0){
      strcat(result,newword);}
    else{
      strcat(result,p);}
    strcat(result," ");
    p=strtok(NULL," \n");}
  printf("Result = %s",result);}
```

74. Remove a particular word

```
#include <stdio.h>
#include <string.h>
int main(){
  char s[200],word[30],result[300]="";
  printf("Enter sentence: ");
```

```
   fgets(s,200,stdin);
   printf("Enter word to remove: ");
   scanf("%s",word);
   char *p=strtok(s," \n");
   while(p!=NULL){
      if(strcmp(p,word)!=0){
         strcat(result,p);
         strcat(result," ");}
      p=strtok(NULL," \n");}
   printf("Result = %s",result);}
```

75. Find number of vowels in each word

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200];
   int count,i;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      count=0;
      for(i=0;p[i]!='\0';i++){
if(p[i]=='a'||p[i]=='e'||p[i]=='i'||p[i]=='o'||p[i]
=='u'||
p[i]=='A'||p[i]=='E'||p[i]=='I'||p[i]=='O'||p[i]=
='U'){
            count++;}}
      printf("%s = %d vowels\n",p,count);
      p=strtok(NULL," \n");}}
```

76. Find word containing maximum number of vowels

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],answer[50];
   int i,count,max=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      count=0;
      for(i=0;p[i]!='\0';i++){
if(p[i]=='a'||p[i]=='e'||p[i]=='i'||p[i]=='o'||p[i]
=='u'||
p[i]=='A'||p[i]=='E'||p[i]=='I'||p[i]=='O'||p[i]=
='U'){
            count++;}}
      if(count>max){
         max=count;
         strcpy(answer,p);}
      p=strtok(NULL," \n");}
   printf("Word = %s",answer);}
```

77. Find word containing maximum number of consonants

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],answer[50];
   int i,count,max=0;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      count=0;
      for(i=0;p[i]!='\0';i++){
if((p[i]>='a'&&p[i]<='z')||(p[i]>='A'&&p[i]<='Z')
){
if(!(p[i]=='a'||p[i]=='e'||p[i]=='i'||p[i]=='o'||p
[i]=='u'||
p[i]=='A'||p[i]=='E'||p[i]=='I'||p[i]=='O'||p[i]=
='U')){
               count++;}}}
      if(count>max){
         max=count;
         strcpy(answer,p);}
      p=strtok(NULL," \n");}
   printf("Word = %s",answer);}
```

78. Arrange words alphabetically

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],words[30][30],temp[30];
   int n=0,i,j;
   printf("Enter sentence: ");
   fgets(s,200,stdin);
   char *p=strtok(s," \n");
   while(p!=NULL){
      strcpy(words[n],p);
      n++;
      p=strtok(NULL," \n");}
   for(i=0;i<n;i++){
      for(j=i+1;j<n;j++){
         if(strcmp(words[i],words[j])>0){
            strcpy(temp,words[i]);
            strcpy(words[i],words[j]);
            strcpy(words[j],temp);}}}
```

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```
    printf("Alphabetical order = ");
    for(i=0;i<n;i++){
        printf("%s ",words[i]);}}
```

79. Find common characters between two strings

```
#include <stdio.h>
int main(){
    char a[100],b[100];
    int i,j,found;
    printf("Enter first string: ");
    fgets(a,100,stdin);
    printf("Enter second string: ");
    fgets(b,100,stdin);
    printf("Common characters = ");
    for(i=0;a[i]!='\0'&&a[i]!='\n';i++){
        found=0;
        for(j=0;b[j]!='\0'&&b[j]!='\n';j++){
            if(a[i]==b[j]){
                found=1;
                break;}}
        if(found){
            printf("%c ",a[i]);}}}
```

80. Find characters present in first string but not second

```
#include <stdio.h>
int main(){
    char a[100],b[100];
    int i,j,found;
    printf("Enter first string: ");
    fgets(a,100,stdin);
    printf("Enter second string: ");
    fgets(b,100,stdin);
    printf("Characters = ");
    for(i=0;a[i]!='\0'&&a[i]!='\n';i++){
        found=0;
        for(j=0;b[j]!='\0'&&b[j]!='\n';j++){
            if(a[i]==b[j]){
                found=1;
                break;}}
        if(!found){
            printf("%c ",a[i]);}}}
```

LEVEL 5: ADVANCED STRING PROBLEMS

81. Longest substring without repeating characters

```
#include <stdio.h>
int main(){
    char s[100];
    int i,j,k,found,max=0,start=0;
    printf("Enter string: ");
    fgets(s,100,stdin);
    for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
        for(j=i;s[j]!='\0'&&s[j]!='\n';j++){
            found=0;
            for(k=i;k<j;k++){
                if(s[k]==s[j]){
                    found=1;
                    break;}}
            if(found){
                break;}
            if(j-i+1>max){
                max=j-i+1;
                start=i;}}}
    printf("Longest substring = ");
    for(i=start;i<start+max;i++){
        printf("%c",s[i]);}}
```

82. Longest substring containing only vowels

```
#include <stdio.h>
int main(){
    char s[100];
    int i,len=0,max=0,start=0,maxstart=0;
    printf("Enter string: ");
    fgets(s,100,stdin);
    for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
if(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]=
='u'||
s[i]=='A'||s[i]=='E'||s[i]=='I'||s[i]=='O'||s[i]=='
U'){
            len++;}
        else{
            if(len>max){
                max=len;
                maxstart=start;}
            len=0;
            start=i+1;}}
    if(len>max){
        max=len;
        maxstart=start;}
    printf("Longest vowel substring = ");
    for(i=maxstart;i<maxstart+max;i++){
        printf("%c",s[i]);}}
```

83. Longest substring containing only consonants

```
#include <stdio.h>
int main(){
    char s[100];
    int i,len=0,max=0,start=0,maxstart=0;
```

```
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
if((s[i]>='a'&&s[i]<='z')||(s[i]>='A'&&s[i]<='Z'))
{
if(!(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]
=='u'||
s[i]=='A'||s[i]=='E'||s[i]=='I'||s[i]=='O'||s[i]=='
U')){
            len++;}
         else{
            if(len>max){
               max=len;
               maxstart=start;}
            len=0;
            start=i+1;}}
      else{
         if(len>max){
            max=len;
            maxstart=start;}
         len=0;
         start=i+1;}}
   if(len>max){
      max=len;
      maxstart=start;}
   printf("Longest consonant substring = ");
   for(i=maxstart;i<maxstart+max;i++){
      printf("%c",s[i]);}}
```

84. Longest substring containing equal numbers of 0s and 1s

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j,zero,one,max=0,start=0;
   printf("Enter binary string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      zero=0;
      one=0;
      for(j=i;s[j]!='\0'&&s[j]!='\n';j++){
         if(s[j]=='0'){
            zero++;}
         else if(s[j]=='1'){
            one++;}
         if(zero==one&&j-i+1>max){
            max=j-i+1;
            start=i;}}}
   printf("Longest substring = ");
   for(i=start;i<start+max;i++){
      printf("%c",s[i]);}}
```

85. Find number of substrings

```
#include <stdio.h>
int main(){
   char s[100];
   int i=0,n=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   while(s[i]!='\0'&&s[i]!='\n'){
      n++;
      i++;}
   printf("Number of substrings =
%d",n*(n+1)/2);}
```

86. Print all possible substrings

```
#include <stdio.h>
int main(){
   char s[100];
   int i,j,k;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      for(j=i;s[j]!='\0'&&s[j]!='\n';j++){
         for(k=i;k<=j;k++){
            printf("%c",s[k]);}
         printf("\n");}}}
```

87. Print all possible subsequences

```
#include <stdio.h>
void subsequence(char s[],char a[],int i,int j){
   if(s[i]=='\0'){
      a[j]='\0';
      printf("%s\n",a);
      return;}
   a[j]=s[i];
   subsequence(s,a,i+1,j+1);
   subsequence(s,a,i+1,j);}
int main(){
   char s[20],a[20];
   printf("Enter string: ");
   fgets(s,20,stdin);
   subsequence(s,a,0,0);}
```

88. Find number of possible subsequences

```
#include <stdio.h>
int main(){
   char s[100];
   int i,n=0;
   long long result=1;
   printf("Enter string: ");
```

```
   fgets(s,100,stdin);
   while(s[n]!='\0'&&s[n]!='\n'){
      n++;}
   for(i=0;i<n;i++){
      result=result*2;}
   printf("Number of subsequences =
%lld",result-1);}
```

89. Find longest common prefix of multiple strings

```
#include <stdio.h>
#include <string.h>
int main(){
   char a[10][100];
   int n,i,j,match=1;
   printf("Enter number of strings: ");
   scanf("%d",&n);
   getchar();
   for(i=0;i<n;i++){
      fgets(a[i],100,stdin);
      a[i][strcspn(a[i],"\n")]='\0';}
   printf("Longest common prefix = ");
   for(j=0;a[0][j]!='\0';j++){
      for(i=1;i<n;i++){
         if(a[i][j]!=a[0][j]){
            match=0;
            break;}}
      if(!match){
         break;}
      printf("%c",a[0][j]);}}
```

90. Find longest common substring between two strings

```
#include <stdio.h>
int main(){
   char a[100],b[100];
   int i,j,k,len,max=0,start=0;
   printf("Enter first string: ");
   fgets(a,100,stdin);
   printf("Enter second string: ");
   fgets(b,100,stdin);
   for(i=0;a[i]!='\0'&&a[i]!='\n';i++){
      for(j=0;b[j]!='\0'&&b[j]!='\n';j++){
         k=0;
while(a[i+k]==b[j+k]&&a[i+k]!='\0'&&a[i+k]!='
\n'){
            k++;}
         if(k>max){
            max=k;
            start=i;}}}
   printf("Longest common substring = ");
   for(i=start;i<start+max;i++){
      printf("%c",a[i]);}}
```

91. Find longest common subsequence

```
#include <stdio.h>
#include <string.h>
int main(){
   char a[50],b[50];
   int dp[50][50]={0};
   int i,j,m,n;
   printf("Enter first string: ");
   fgets(a,50,stdin);
   printf("Enter second string: ");
   fgets(b,50,stdin);
   a[strcspn(a,"\n")]='\0';
   b[strcspn(b,"\n")]='\0';
   m=strlen(a);
   n=strlen(b);
   for(i=1;i<=m;i++){
      for(j=1;j<=n;j++){
         if(a[i-1]==b[j-1]){
            dp[i][j]=dp[i-1][j-1]+1;}
         else if(dp[i-1][j]>dp[i][j-1]){
            dp[i][j]=dp[i-1][j];}
         else{
            dp[i][j]=dp[i][j-1];}}}
   printf("LCS length = %d\n",dp[m][n]);}
```

92. Check whether two strings are one edit away

```
#include <stdio.h>
#include <string.h>
int main(){
   char a[100],b[100];
   int i=0,j=0,diff=0;
   printf("Enter first string: ");
   fgets(a,100,stdin);
   printf("Enter second string: ");
   fgets(b,100,stdin);
   a[strcspn(a,"\n")]='\0';
   b[strcspn(b,"\n")]='\0';
if(strlen(a)>strlen(b)+1||strlen(b)>strlen(a)+1)
{
      printf("Not one edit away");
      return 0;}
   while(a[i]!='\0'&&b[j]!='\0'){
      if(a[i]!=b[j]){
         diff++;
         if(diff>1){
```

```
            break;}
        if(strlen(a)>strlen(b)){
            i++;}
        else if(strlen(b)>strlen(a)){
            j++;}
        else{
            i++;
            j++;
            continue;}}
      else{
        i++;
        j++;}}
   if(diff<=1){
      printf("One edit away");}
   else{
      printf("Not one edit away");}}
```

93. Minimum characters required to make a string palindrome

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[100];
   int i,j,count=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   s[strcspn(s,"\n")]='\0';
   i=0;
   j=strlen(s)-1;
   while(i<j){
      if(s[i]==s[j]){
         i++;
         j--;}
      else{
         count++;
         j--;}}
   printf("Minimum characters = %d",count);}
```

94. Minimum deletions to make palindrome

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[100];
   int dp[100][100]={0};
   int i,j,n;
   printf("Enter string: ");
   fgets(s,100,stdin);
   s[strcspn(s,"\n")]='\0';
   n=strlen(s);
   for(i=1;i<n;i++){
      for(j=0;j<n-i;j++){
         if(s[j]==s[j+i]){
            dp[j][j+i]=dp[j+1][j+i-1];}
         else{
            dp[j][j+i]=1+(
               dp[j+1][j+i]<dp[j][j+i-1]?
               dp[j+1][j+i]:dp[j][j+i-1]);}}}
   printf("Minimum deletions = %d",dp[0][n-
1]);}
```

95. Minimum insertions to make palindrome

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[100];
   int dp[100][100]={0};
   int i,j,n;
   printf("Enter string: ");
   fgets(s,100,stdin);
   s[strcspn(s,"\n")]='\0';
   n=strlen(s);
   for(i=1;i<n;i++){
      for(j=0;j<n-i;j++){
         if(s[j]==s[j+i]){
            dp[j][j+i]=dp[j+1][j+i-1];}
         else{
            dp[j][j+i]=1+(
               dp[j+1][j+i]<dp[j][j+i-1]?
               dp[j+1][j+i]:dp[j][j+i-1]);}}}
   printf("Minimum insertions = %d",dp[0][n-
1]);}
```

96. Check whether string can be rearranged to form palindrome

```
#include <stdio.h>
int main(){
   char s[100];
   int count[256]={0};
   int i,odd=0;
   printf("Enter string: ");
   fgets(s,100,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      count[(unsigned char)s[i]]++;}
   for(i=0;i<256;i++){
      if(count[i]%2!=0){
         odd++;}}
   if(odd<=1){
      printf("Can form palindrome");}
   else{
      printf("Cannot form palindrome");}}
```

97. Find smallest window containing all characters of another string

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],p[100];
   int need[256]={0},have[256]={0};
   int i,left=0,count=0,start=0,min=1000;
   printf("Enter main string: ");
   fgets(s,200,stdin);
   printf("Enter required string: ");
   fgets(p,100,stdin);
   s[strcspn(s,"\n")]='\0';
   p[strcspn(p,"\n")]='\0';
   for(i=0;p[i]!='\0';i++){
      need[(unsigned char)p[i]]++;}
   for(i=0;s[i]!='\0';i++){
      have[(unsigned char)s[i]]++;
      if(have[(unsigned
char)s[i]]<=need[(unsigned char)s[i]]){
         count++;}
      while(count==strlen(p)){
         if(i-left+1<min){
            min=i-left+1;
            start=left;}
         have[(unsigned char)s[left]]--;
         if(have[(unsigned
char)s[left]]<need[(unsigned char)s[left]]){
            count--;}
         left++;}}
   if(min==1000){
      printf("Window not found");}
   else{
      printf("Smallest window = ");
      for(i=start;i<start+min;i++){
         printf("%c",s[i]);}}}
```

98. Compress a string using counts

Example: aaabbc → a3b2c1

```
#include <stdio.h>
int main(){
   char s[100];
   int i=0,count;
   printf("Enter string: ");
   fgets(s,100,stdin);
   while(s[i]!='\0'&&s[i]!='\n'){
      count=1;
      while(s[i]==s[i+count]){
         count++;}
      printf("%c%d",s[i],count);
      i=i+count;}}
```

99. Implement own strstr() to find substring

```
#include <stdio.h>
int main(){
   char s[200],sub[50];
   int i,j,found=0;
   printf("Enter string: ");
   fgets(s,200,stdin);
   printf("Enter substring: ");
   fgets(sub,50,stdin);
   for(i=0;s[i]!='\0'&&s[i]!='\n';i++){
      j=0;
while(sub[j]!='\0'&&sub[j]!='\n'&&s[i+j]==sub[
j]){
         j++;}
      if(sub[j]=='\0'||sub[j]=='\n'){
         found=1;
         printf("Substring found at position
%d",i);
         break;}}
   if(!found){
      printf("Substring not found");}}
```

100. Complete string utility menu program

```
#include <stdio.h>
#include <string.h>
int main(){
   char s[200],a[100],b[100],ch;
   int choice,i,j,count;
   printf("Enter string: ");
   fgets(s,200,stdin);
   s[strcspn(s,"\n")]='\0';
   while(1){
      printf("\n1.Length");
      printf("\n2.Copy");
      printf("\n3.Compare");
      printf("\n4.Concatenate");
      printf("\n5.Reverse");
      printf("\n6.Palindrome");
      printf("\n7.Frequency");
      printf("\n8.Substring search");
      printf("\n9.Replace character");
      printf("\n10.Exit");
      printf("\nEnter choice: ");
      scanf("%d",&choice);
      getchar();
```

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```
if(choice==1){
   printf("Length = %d",(int)strlen(s));}
else if(choice==2){
   strcpy(a,s);
   printf("Copied string = %s",a);}
else if(choice==3){
   printf("Enter second string: ");
   fgets(a,100,stdin);
   a[strcspn(a,"\n")]='\0';
   if(strcmp(s,a)==0){
      printf("Equal");}
   else{
      printf("Not equal");}}
else if(choice==4){
   printf("Enter second string: ");
   fgets(a,100,stdin);
   a[strcspn(a,"\n")]='\0';
   strcat(s,a);
   printf("Result = %s",s);}
else if(choice==5){
   i=0;
   j=strlen(s)-1;
   while(i<j){
      ch=s[i];
      s[i]=s[j];
      s[j]=ch;
      i++;
      j--;}
   printf("Reverse = %s",s);}
else if(choice==6){
   i=0;
   j=strlen(s)-1;
   count=1;
   while(i<j){
      if(s[i]!=s[j]){
         count=0;
         break;}
      i++;
      j--;}
   if(count){
      printf("Palindrome");}
   else{
      printf("Not palindrome");}}
else if(choice==7){
   int f[256]={0};
   for(i=0;s[i]!='\0';i++){
      f[(unsigned char)s[i]]++;}
   for(i=0;i<256;i++){
      if(f[i]>0){
         printf("%c = %d\n",i,f[i]);}}
}
else if(choice==8){
   printf("Enter substring: ");
   fgets(a,100,stdin);
   a[strcspn(a,"\n")]='\0';
   if(strstr(s,a)!=NULL){
      printf("Substring found");}
   else{
      printf("Substring not found");}}
else if(choice==9){
   printf("Enter character: ");
   scanf("%c",&ch);
   getchar();
   printf("Enter new character: ");
   scanf("%c",&b[0]);
   getchar();
   for(i=0;s[i]!='\0';i++){
      if(s[i]==ch){
         s[i]=b[0];}}
   printf("Result = %s",s);}
else if(choice==10){
   break;}
else{
   printf("Invalid choice");}}}
```

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